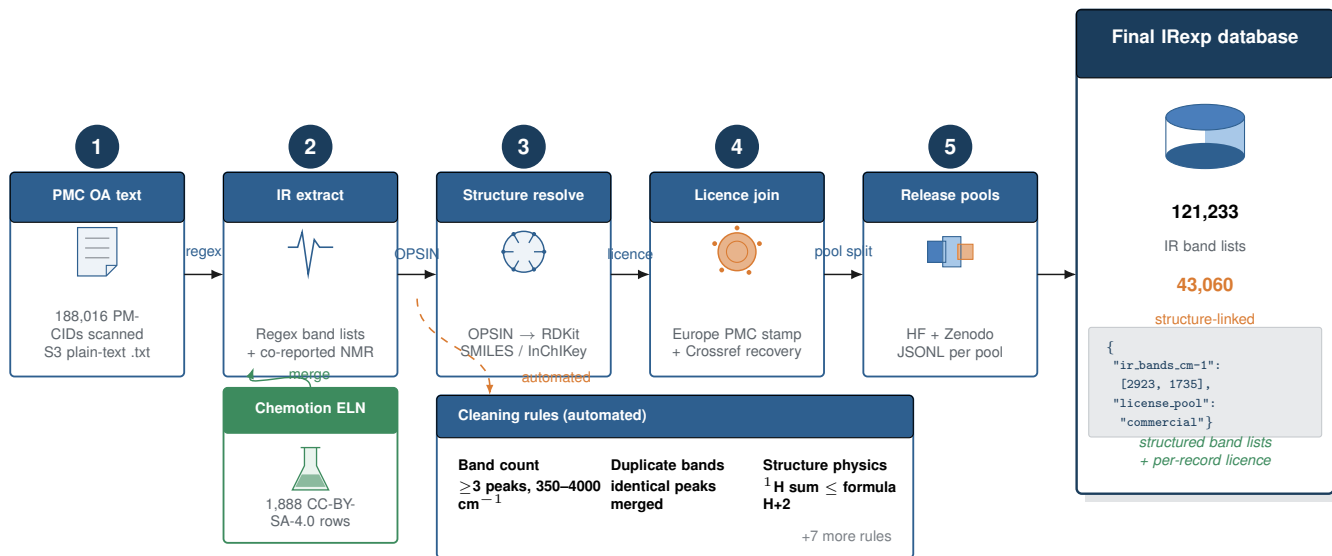


A Extraction and quality-control workflow



B Illustrative automated QC rejections

IR (KBr): 3021, 2923, 2854 cm^{-1}

^1H NMR (480 MHz, CDCl_3): δ 7.25 (2H), 6.80 (2H)

Rejected — band count 3 < minimum gate after de-duplication



IR (neat): 3200, 2958, 4180, 1520 cm^{-1}

^{13}C NMR: δ 171.2, 133.5, 128.1, 115.4

Rejected — band at 4180 cm^{-1} outside 350–4000 cm^{-1} window



IR (film): 3050, 2920, 1650, 1450 cm^{-1}

^1H NMR δ 8.10 (5H), 7.45 (5H), 6.20 (10H); formula $\text{C}_9\text{H}_{11}\text{NO}_2$

Quarantined — ^1H integral sum 20H > formula H+2 (12H)



Automated rule-based filter examples — not human-audited; see Panel-A cleaning rules.